

Research paper

Instability zones in the dynamics of a quantum mechanical quasiperiodic parametric oscillator



Subhadip Biswas^{a,*}, Pratyusha Chowdhury^b, Jayanta K Bhattacharjee^{c,*}

^a Department of Physics and Astronomy, University of Sheffield, Sheffield S3 7RH, UK

^b Department of Physics, Indian Institute of Technology Guwahati, Guwahati, Assam 781039, India

^c Department of Theoretical Physics, Indian Association for the Cultivation of Science, Jadavpur, Kolkata 700032, India

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ABSTRACT

Quasiperiodically driven quantum parametric oscillators have been the subject of several recent investigations. Here we show that for such oscillators, the instability zones of the mean position and variance (alternatively the mean energy) for a time developing wave packet are identical for the strongest resonance in the three-dimensional parameter space of the quasiperiodic modulation as it is for the two-dimensional parameter space of the periodic modulations.

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1. Introduction

The quantum parametric oscillator has drawn a fair amount of attention because of its relevance in the physics of ion traps [1] as well as in the study of the longtime properties of periodically driven quantum system [2,3]. A particular issue about the quantum parametric oscillators is the existence of regions of instability in the plane spanned by amplitude and frequency of modulation. The mean position of an initially prepared wave packet diverges in certain regions of the plane. More importantly, the variance and hence the energy of the oscillator increases indefinitely in certain regions of the plane. The mean position diverging implies that the particle escapes from the trap. The average energy increasing indefinitely is also undesirable. It is consequently important to know the zones where these divergences occur. Recent work [4–6] has established quite clearly that the variance and the mean diverge in the same regions of the frequency-amplitude plane leading to the situation that if the particle is trapped its energy will be finite.

While references [4] and [5] have dealt with the variance, it should be noted that reference [6] has dealt with a more physical quantity which is the energy expectation value and shown that the region of divergence of mean position and the average energy are the same. We show the connection between the variance and the average energy in Section 2.

Over the last few years, there has been a fair amount of discussion [7–11] on quasiperiodically driven quantum systems. Some of it is related to the efforts to generalize Floquet's theorem to quasiperiodic case [12–14]. In this work, we address the question of whether the instability regions are identical for the mean position and the variance (and hence the energy) for the quasiperiodically driven system as well. To answer this question, we exploit the fact that the system is a simple harmonic

* Corresponding authors.

E-mail addresses: sbiswas2@sheffield.ac.uk (S. Biswas), jayanta.bhattacharjee@gmail.com (J.K. Bhattacharjee).

oscillator and hence the dynamics of the mean position and the variance (and for that matter all the higher moments) are exactly known. We look at the worst case scenario - the situation where one of the driving frequencies is in the primary parametric situation i.e. the driving frequency is twice the natural frequency and the other frequency which makes the drive quasiperiodic is only slightly detuned leading to the most spectacular instability zones. We show numerically and through a perturbative analysis that the instability zones for the mean and variance are identical in this situation.

The layout of the paper is as follows: In [Section 2](#), we recall the dynamics of the mean position and variance of the quantum parametric oscillator for an arbitrary forcing function $f(t)$. If the natural frequency of the physical oscillator is ω ($T = \frac{2\pi}{\omega}$ is the time period), then the oscillator for the mean position has the same natural frequency ω and satisfies the usual *Mathieu* equation. The oscillator describing the dynamics of the variance will be seen to have a natural frequency of 2ω (natural frequency is the frequency of the autonomous system) and will be described by a third-order non-autonomous linear differential equation. We then explain the rationale behind working with a definite form of $f(t)$. In [Section 3](#) we discuss the case around $T = \frac{2\pi}{\omega}$, i.e. twice the natural frequency. Here, a cancellation prevents the occurrence of any instability zones. We show this from a perturbative calculation which is supported by numerical calculation of the stability boundary. In [Section 4](#), we investigate the region near $T = \frac{\pi}{\omega}$ for the dynamics of the variance on the basis of perturbation theory and it is here that one finds a variety of instability zones and they agree exactly with the corresponding zones for the response near twice the natural period for the dynamics of the mean. In [Section 5](#), we present complete numerical solutions of the governing equations to show that the instability zones of the mean and variance precisely overlap. We conclude with a brief summary in [Section 6](#).

2. The equation of motion

The one dimensional parametric oscillator is governed by the Hamiltonian $\mathcal{H}$ given by

$$\mathcal{H} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2(1 + \epsilon f(t)), \quad (2.1)$$

where m is the mass of the oscillator, p its momentum, x its coordinate and ω the frequency of the vibration. The restoring force is modified by the time dependent function $f(t)$ which has an amplitude ϵ . In this work we will consider a quasiperiodic $f(t)$ which can be written as,

$$f(t) = \cos(\Omega t) + \mu \cos(1 + \epsilon \Delta)\Omega t. \quad (2.2a)$$

This is a special case (in many ways the most interesting as we show at the end of this section) of the general form

$$f(t) = \cos(\Omega t) + \mu \cos(1 + \alpha)\Omega t, \quad (2.2b)$$

where α is an irrational number.

The *Schrödinger* equation, corresponding to the Hamiltonian of [Eq. \(2.1\)](#) is $i\hbar \frac{\partial \Psi}{\partial t} = \mathcal{H}\Psi$, where $\Psi(x, t)$ is the space and time dependent wave function of the system. For any operator $\mathcal{O}$ we can write down the *Heisenberg* equation

$$i\hbar \frac{d\mathcal{O}}{dt} = i\hbar \frac{\partial \mathcal{O}}{\partial t} + [\mathcal{O}, \mathcal{H}], \quad (2.3)$$

where the first term on the right hand side is nonvanishing only if the operator $\mathcal{O}$ is explicitly time dependent and $[\mathcal{O}, \mathcal{H}]$ is the commutator $\mathcal{O}\mathcal{H} - \mathcal{H}\mathcal{O}$. Taking the expectation value of the operators in the above equation in any state $\Psi(x, t)$, we have

$$i\hbar \frac{d}{dt} \langle \mathcal{O} \rangle = i\hbar \left\langle \frac{\partial \mathcal{O}}{\partial t} \right\rangle + \langle [\mathcal{O}, \mathcal{H}] \rangle, \quad (2.4)$$

where $\langle \dots \rangle = \int dx \Psi^*(x, t) (\dots) \Psi(x, t)$. As mentioned in the introduction, our focus here will be on the expectation value of the position operator which in this case actually follows the classical equation of motion (this happens for all quadratic Hamiltonians) and the variance which is a purely quantum mechanical object and has no classical analogue except when one considers a bunch of initial conditions in classical dynamics [\[15\]](#).

The dynamics of $\langle x \rangle$, the expectation value of the position operator and the dynamics of the variance $V = \langle x^2 \rangle - \langle x \rangle^2$ have been found for an arbitrary $f(t)$ in Ref. [\[5\]](#). Governing equations are as follows,

$$\frac{d^2}{dt^2} \langle x \rangle + \omega^2 [1 + \epsilon \cos(\Omega t) + \epsilon \mu \cos(1 + \epsilon \Delta)\Omega t] \langle x \rangle = 0, \quad (2.5)$$

$$\frac{d^3 V}{dt^3} + 4\omega^2(1 + \epsilon f(t)) \frac{dV}{dt} + 2\omega^2 \epsilon \dot{f} V = 0. \quad (2.6)$$

The dynamics of the variance and with the form of $f(t)$ as given in [Eq. \(2.2a\)](#) is

$$\begin{aligned} \frac{d^3 V}{dt^3} + 4\omega^2 \frac{dV}{dt} + 4\epsilon \omega^2 [\cos(\Omega t) + \mu \cos \Omega(1 + \epsilon \Delta)t] \frac{dV}{dt} \\ - 2\omega^2 \epsilon \Omega [\sin(\Omega t) + \mu(1 + \epsilon \Delta) \sin \Omega(1 + \epsilon \Delta)t] V = 0. \end{aligned} \quad (2.7)$$

The fact that the behaviour of the classical system is striking was realized in the Ref. [16]. The dynamics of the mean as shown in Eq. (2.5) is just the classical dynamics and has been extensively studied [16–19]. The dynamics of the variance as shown in Eq. (2.7) has been investigated in Biswas et al. [4], Biswas and Bhattacharjee [5] for $\mu = 0$ (periodic modulation).

We note that successive applications of Eq. (2.4) yields,

$$\frac{d}{dt}\langle E \rangle = \frac{d}{dt} \left\langle \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 (1 + \epsilon f(t)) \right\rangle = -\frac{1}{2}m\omega^2 \epsilon \langle x^2 \rangle \frac{d}{dt} f(t), \quad (2.8)$$

hence the instability zones of $\langle E \rangle$ are linked to these of $\langle x^2 \rangle$. Our aim here will be to see how different the quasiperiodic perturbation is from the periodic one given that for the mean, the changes in the instability zone are drastic for $\mu \neq 0$ (the quasiperiodic case).

The technique of working with Eqs. (2.2a) and (2.2b) have been carefully dealt with in Refs. [16–19], only in the case where Δ is irrational and ϵ is very small. Here we point out the only feature that has not been explained in detail in those papers. The authors of [16–19] do not consider such large values of $\epsilon\Delta$ and for a very good reason which we will try to elaborate. We begin with a brief recall of Floquet theory.

A n^{th} -order differential equation can always be cast as a set of n coupled first order differential equations. As a trivial example Eq. (2.5) can be written as the as the set of two equations:

$$\begin{aligned} \frac{d}{dt} \langle x \rangle &= y, \\ \frac{dy}{dt} &= -\omega^2 [1 + \epsilon f(t)] \langle x \rangle. \end{aligned}$$

We can think of $\langle x \rangle$ and y as the two components of a vector $\vec{X}$ and Eq. (2.5) can be written in the form

$$\dot{X}_i = \sum_{j=1}^2 M_{ij} X_j, \quad (2.9)$$

where M_{ij} is the 2×2 matrix $\begin{pmatrix} 0 & 1 \\ -\omega^2[1 + \epsilon f(t)] & 0 \end{pmatrix}$. In a similar fashion Eq. (2.7) can be cast in the form of Eq. (2.9) with M a 3×3 matrix. Floquet's theorem deals with the situation where we have Eq. (2.9) with M as a periodic function with minimal period T i.e. T is the smallest positive number for which $M(t) = M(t + T)$. The theorem states (see e.g. Ref. [20]) that if the vector $\vec{X} \{X_i, i = 1 \dots n\}$ follows the dynamics $\dot{X}_i = M_{ij} X_j$ with M_{ij} a $n \times n$ matrix of minimal period T , then there must exist at least one solution $X_i = \mathcal{X}_i(t)$ such that

$$\mathcal{X}_i(t + T) = \zeta \mathcal{X}_i(t). \quad (2.10)$$

The possible numbers ζ arise as the eigenvalues of the matrix E_{ij} which connects a solution matrix (each solution vector ϕ_i has n -components and there are n linearly independent solutions) $\phi_{ij}(t)$ to the solution matrix $\phi_{ij}(t + T)$. Periodic solutions of Eq. (2.9) corresponds to ζ having the form $e^{2\pi i/m}$ where m is an integer. The n -eigenvalues (including possible repeats of a particular eigenvalue) $\zeta_1, \zeta_2, \dots, \zeta_n$ satisfy

$$\zeta_1 \zeta_2 \dots \zeta_n = e^{\int_0^T \text{Tr}[M(s)] ds}. \quad (2.11)$$

We see from Eq. (2.10) that a ζ of the form $e^{2\pi i/m}$ corresponds to a solution of periodic mT . Instabilities arise when the magnitude of the generally complex number become greater than unity. For the Mathieu equation $\ddot{x} + \omega^2[1 + \epsilon f(t)] = 0$, these instability zones are very well known. The matrix M of Eq. (2.9) has period $T = \frac{2\pi}{\omega}$. The trace of M is zero in this case and the two eigenvalues ζ_1 and ζ_2 satisfy $\zeta_1 \zeta_2 = 1$. Periodic solutions arise when the integer $m = 1$ or 2 i.e. $\zeta = \pm 1$. For $\zeta = -1$, the periodicity is $2T$ and for $\zeta = 1$, the periodicity is T . The natural frequency of the system (i.e. frequency when $\epsilon = 0$) is ω and a possible resonant situation (leading to instability) arises with $2T = \frac{2\pi}{\omega}$ or $T = \frac{2\pi}{\omega}$ i.e. $\omega T = \pi$ or 2π .

Every irrational number has an infinite continued fraction expansion and stopping it at some point leads to a rational number approximation. What happens to the stability zones as one works with the rational approximants and approaches the irrational number? We illustrate this limiting procedure with the golden ratio $\frac{\sqrt{5}-1}{2}$. The successive rational approximants to the irrational $\alpha = \epsilon\Delta = \frac{\sqrt{5}-1}{2}$ are $\alpha = \frac{1}{2}, \frac{2}{3}, \frac{3}{5}, \frac{5}{8}, \frac{8}{13}, \dots$ converging very quickly to $\alpha = 0.618\dots$ For each of the rational approximations, $f(t)$ is a periodic function with a period T that increases as the approximants converge to the irrational number.

We begin by applying Floquet theory. For $\alpha = \frac{1}{2}$, $f(t)$ has a period $\frac{4\pi}{\Omega}$, for $\alpha = \frac{2}{3}$ the period is $\frac{6\pi}{\Omega}$ and so on. Floquet theory says that for $\alpha = \frac{1}{2}$, the frequencies around which one looks for instability in the response are $\frac{n\Omega}{4}$ ($n = 1, 2, 3, \dots$), while for $\alpha = \frac{2}{3}$ one looks for instability around $\frac{n\Omega}{6}$. The basic frequency ($\frac{\Omega}{4}, \frac{\Omega}{6}, \dots$) goes on decreasing and eventually tends to zero as α becomes $\frac{\sqrt{5}-1}{2}$. The fact that one is required to look for resonances near $\frac{n\Omega}{4}$, does not mean that there will be an instability zone for all n . For $n = 1$, it is straightforward to see by repeating the steps shown in Section 3, that there is no instability zone around $\frac{\Omega}{4}$. There is a periodic orbit of period $\frac{8\pi}{\Omega}$ (Floquet's theorem) but it exists in the $\epsilon - \omega$ plane along the curve $\delta = -\frac{\epsilon^2}{\Omega^2}$ where $\omega^2 = \frac{\Omega^2}{16} + \delta$ for small values of ϵ . The primary instability zones are around $\frac{\Omega}{2}$ (for $n = 2$) and $\frac{3\Omega}{4}$

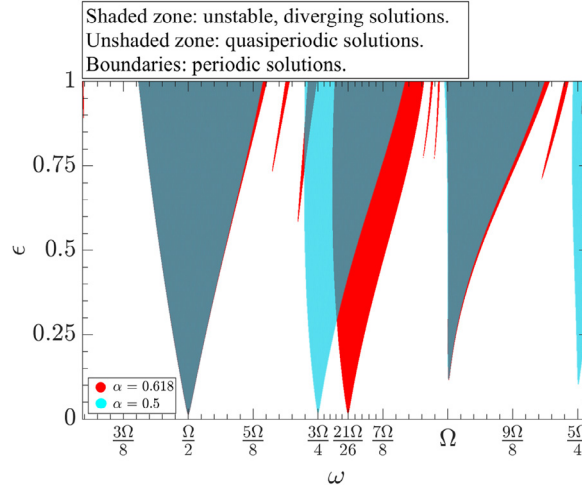


Fig. 1. Numerical solution of Eq. (2.5) with $f(t)$ from Eq. (2.2b) with $\mu = 0.5$, $\alpha = 0.5$ and $\alpha = \frac{8}{13}$, $\Omega = 2\pi$ in the $\epsilon - \omega$ parameter space. Coloured points are the unstable solution and white region corresponds to stable solution of the Mathieu equation. The instability zones of V obtained from Eq. (2.7) with the same set of parameters mentioned above yield same region of the above figure.

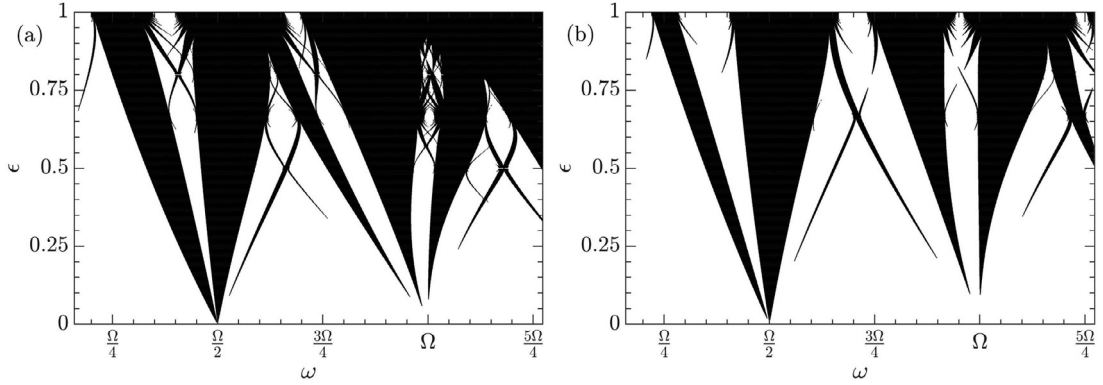


Fig. 2. Numerical solution of Eq. (2.5) for two different μ , (a) $\mu = 1.0$ and (b) $\mu = 0.5$, with fixed values of $\Delta = -0.5$, $\Omega = 2\pi$ in the $\epsilon - \omega$ parameter space. Black points are the unstable solution and white region corresponds to stable solution of the Mathieu equation. The boundaries are now far more complicated than those shown in Fig. 1.

(for $n = 3$), followed by $n = 4$. This picture hardly changes for the next approximant except that the significant instability zones are around $\frac{n_2\Omega}{6}$ with $n = 3$ and $n = 5$, followed by $n = 6$. In the limiting (quasiperiodic) situation the significant zones are around $\frac{n_1\Omega}{2}$ ($n_1 = 1, 2$) and $n_2\frac{\sqrt{5}+1}{2}$ ($n_2 = 1, 2$).

The numerically obtained instability zones corroborate the above statements. The point of the present paper is that the instability zones of the variance (as obtained from the numerical or perturbative treatment of Eq. (2.7)) yield the identical instability zones. We show this numerically in Fig. 1, where we have (to prevent crowding) shown the two cases of $\alpha = \frac{1}{2}$ and $\alpha = \frac{8}{13}$. The case $\alpha = \frac{8}{13}$ is virtually identical to $\alpha = 0.618$ and not very different from the first approximant $\alpha = \frac{1}{2}$. We repeat that the plots of instability zones from Eqs. (2.5) and (2.7) coincide in all cases. This is not a surprise as it is very similar to the cases treated in Ref. [5] in spite of the drive being quasiperiodic. What one sees in Fig. 1, is simply the results of Ref. [5] for two separate driving frequencies Ω and $\frac{\sqrt{5}+1}{2}\Omega$.

What we learn from above is that when the frequencies Ω and $\Omega(1 + \epsilon\Delta)$ of Eq. (2.2a) are well separated, the instability zones which are most prominent when originating from $\omega = \frac{\Omega}{2}$ or $\frac{\Omega}{2}(1 + \epsilon\Delta)$ are also well separated.

Only if $\epsilon\Delta$ is made very small, the instability zones overlap and can produce an immense amount of fine structures as shown in Fig. 2. This is what was noted by Rand et al. [16] and thus it is important to focus only on the situation $\epsilon\Delta < 1$. Consequently it is important to establish that the two immensely complicated instability zones agree exactly for the mean (Eq. (2.5)) and variance (Eq. (2.7)) both in a direct numerical investigation and in a perturbative Krylov-Bogoliubov analysis. The perturbative results for the instability zones from the Krylov-Bogoliubov technique will be shown in Sections 3 and 4. The full numerical results will be shown in Section 5. For purely periodic $f(t)$ of period $T = \frac{2\pi}{\Omega}$ in Eq. (2.6), the periodic orbit can occur with periods $3T$, $2T$ and T . This is because Eq. (2.6) can be written as third order traceless dynamical system having the structure $\dot{X}_i = \sum_{j=1}^3 M_{ij}(t)X_j$, where $M_{ij}(t)$ has zero trace and period T .

By inspection, we see from a harmonic balance that at lowest order there will be no instability zone around the orbit of period $3T$. This can be understood by writing the resonance condition as $2\omega = \frac{2\pi}{3T} = \frac{\Omega}{3}$. Instability would imply the possibility of a solution under the condition i.e. all terms of Eq. (2.7) must support a solution $V(t) = A \cos \frac{\Omega t}{3}$. The third and fourth terms of Eq. (2.7) are of the form $\cos \frac{2\Omega t}{3}$ and hence do not support $\cos \frac{\Omega t}{3}$ coming from the first two terms. Hence there is no harmonic balance unless A is trivially zero. In the next section, we begin by looking for orbits of period $2T$.

3. In the vicinity of orbits of period $2T$: perturbation theory

In this section, we describe in details the basic analytic technique following Ref. [5] that will be used in this paper. We will do so for Eq. (2.7) but the identical procedure works for Eq. (2.5). The natural frequency of the oscillator Eq. (2.7) is 2ω and hence if we have 2ω to be in the vicinity of $\frac{\Omega}{2}$ where $\Omega = \frac{2\pi}{T}$, we can expect a strong resonance in the system from our discussion in Section 2. Accordingly, we set the frequency ω near $\frac{\Omega}{4}$ and write

$$\omega = \frac{\Omega}{4} + \delta, \quad (3.1)$$

where $\delta \ll \mathcal{O}(1)$ and can be expressed in a power series in ϵ as

$$\delta = \delta_1 \epsilon + \delta_2 \epsilon^2 + \dots \quad (3.2)$$

To $\mathcal{O}(\epsilon)$, we can now rewrite Eq. (2.7) as

$$\begin{aligned} \frac{d^3 V}{dt^3} + \frac{\Omega^2}{4} \frac{dV}{dt} + 2\Omega \epsilon \delta_1 \frac{dV}{dt} = & -\frac{\epsilon \Omega^2}{4} [\cos(\Omega t) + \mu \cos \Omega(1 + \epsilon \Delta)t] \frac{dV}{dt} + \frac{\epsilon \Omega^3}{8} [\sin(\Omega t) \\ & + \mu(1 + \epsilon \Delta) \sin \Omega(1 + \epsilon \Delta)t] V. \end{aligned} \quad (3.3)$$

We propose to use the *Krylov-Bogoliubov* technique to explore the dynamics $V(t)$ for ω in the vicinity of $\frac{\Omega}{4}$. Accordingly, we note that for $\epsilon = 0$, the solution of Eq. (3.3) is

$$V_0 = A_0 + A \cos\left(\frac{\Omega t}{2}\right) + B \sin\left(\frac{\Omega t}{2}\right), \quad (3.4)$$

where A_0 , A and B are constants. To take the $\mathcal{O}(\epsilon)$ term into account, we assume that A_0 , A and B will become slowly varying functions of time with the structure of the solution at the leading order unchanged. Slowly varying implies that as we take derivatives of A_0 , A and B , only the first order derivative will be retained, while the higher order ones will be ignored. We thus try a solution for $V(t)$ of the form (correct to $\mathcal{O}(\epsilon)$)

$$V(t) = A_0(t) + A(t) \cos\left(\frac{\Omega t}{2}\right) + B(t) \sin\left(\frac{\Omega t}{2}\right). \quad (3.5)$$

The *Krylov-Bogoliubov* procedure requires arriving at the dynamics of A_0 , A and B . Inserting the above form of $V(t)$ in Eq. (3.3), noting that (keeping in mind that derivatives of A_0 , A and B can only be of the first order)

$$\dot{V}(t) = \dot{A}_0(t) + \left(\dot{A} + B \frac{\Omega}{2}\right) \cos\left(\frac{\Omega t}{2}\right) + \left(\dot{B} - A \frac{\Omega}{2}\right) \sin\left(\frac{\Omega t}{2}\right). \quad (3.6a)$$

$$\ddot{V}(t) = \left(\Omega \dot{B} - A \frac{\Omega^2}{4}\right) \cos\left(\frac{\Omega t}{2}\right) - \left(\Omega \dot{A} + B \frac{\Omega^2}{4}\right) \sin\left(\frac{\Omega t}{2}\right). \quad (3.6b)$$

$$\ddot{\ddot{V}}(t) = -\left(\frac{3\Omega^2}{4} \dot{A} + B \frac{\Omega^3}{8}\right) \cos\left(\frac{\Omega t}{2}\right) - \left(\frac{3\Omega^2}{4} \dot{B} - A \frac{\Omega^3}{8}\right) \sin\left(\frac{\Omega t}{2}\right). \quad (3.6c)$$

We simplify anticipating that $\dot{A}_0$, $\dot{A}$ and $\dot{B}$ will be proportional to ϵ (slowly varying implies the derivative is small) and accordingly ignore all terms which are like $\epsilon \dot{A}_0$, $\epsilon \dot{A}$ and $\epsilon \dot{B}$ etc. Use of standard trigonometric identities lead to Eq. (3.3) taking

the form (correct to $\mathcal{O}(\epsilon)$ on the right hand side)

$$\begin{aligned} \frac{\Omega^2}{4}\dot{A}_0 - \frac{\Omega^2}{2}\dot{A}\cos\left(\frac{\Omega t}{2}\right) - \frac{\Omega^2}{2}\dot{B}\sin\left(\frac{\Omega t}{2}\right) &= \epsilon\Omega^2\delta_1\left[A\sin\left(\frac{\Omega t}{2}\right) - B\cos\left(\frac{\Omega t}{2}\right)\right] \\ + \frac{\epsilon\Omega^3}{8}\left[A_0\sin(\Omega t) + A\sin\left(\frac{3\Omega t}{2}\right) - B\cos\left(\frac{3\Omega t}{2}\right)\right. \\ &\quad \left. + \mu A\sin\left(\frac{3\Omega}{2} + \epsilon\Omega\Delta\right)t - \mu B\cos\left(\frac{3\Omega}{2} + \epsilon\Omega\Delta\right)t\right]. \end{aligned} \quad (3.7)$$

Matching the coefficients of similar trigonometric terms on the left and right sides of Eq. (3.5), we have

$$\dot{A}_0 = 0. \quad (3.8a)$$

$$\dot{A} = 2\delta_1 B\epsilon. \quad (3.8b)$$

$$\dot{B} = -2\delta_1 A\epsilon. \quad (3.8c)$$

Using the above equations $A(t)$ is obtained from (identical equation holds for $B(t)$),

$$\ddot{A} + 4\delta_1^2\epsilon^2 A = 0. \quad (3.9)$$

If we look at the $\omega - \epsilon$ plane, at the point $\omega = \frac{\Omega}{4}$, the trajectory is periodic along the vertical line. If we move away a small distance δ_1 from $\omega = \frac{\Omega}{4}$, the trajectory is quasiperiodic with the frequencies $\frac{\Omega}{2} \pm 2\delta_1\epsilon$, as seen from Eqs. (3.5) and (3.9). Thus there is no instability zone of Eq. (2.7) around $\omega = \frac{\Omega}{4}$ to $\mathcal{O}(\epsilon)$. This is exactly what had happened for the purely periodic case of $\mu = 0$ and no qualitative change occurs for $\mu \neq 0$.

4. In the vicinity of orbits of period T: perturbation theory

In this case, our natural frequency 2ω in Eq. (2.7) needs to be close to the forcing frequency Ω and hence

$$\omega = \frac{\Omega}{2} + \delta = \frac{\Omega}{2} + \epsilon\delta_1 + \mathcal{O}(\epsilon^2). \quad (4.1)$$

We first recall the results for the mean position (Eq. (2.5)), where we try the solution (Krylov-Bogoliubov)

$$\langle x \rangle = A(t)\cos\left(\frac{\Omega t}{2}\right) + B(t)\sin\left(\frac{\Omega t}{2}\right), \quad (4.2)$$

where $A(t)$ and $B(t)$ are slowly varying amplitudes. Calculations identical to that shown in Section 3 lead to the dynamics (with $\tau = \Delta\epsilon t\Omega$)

$$\Delta\frac{dA}{d\tau} = \left(\frac{\delta_1}{\Omega} - \frac{1}{8}\right)B - \frac{\mu A}{8}\sin\tau - \frac{\mu B}{8}\cos\tau. \quad (4.3a)$$

$$\Delta\frac{dB}{d\tau} = -\left(\frac{\delta_1}{\Omega} + \frac{1}{8}\right)A - \frac{\mu A}{8}\cos\tau + \frac{\mu B}{8}\sin\tau. \quad (4.3b)$$

These flows agree exactly with the results following from the two-time scale technique of Rand et al. [16]. For $\mu = 0$, the results are in exact agreement with those known for the Mathieu equation and these ($\delta_1 = \pm\frac{\Omega}{8}$) are the boundaries which were obtained in the previous work [5] as the primary resonance. On the lines $\delta_1 = \pm\frac{\Omega}{8}$, A and B are constants and hence Eq. (4.2) gives a periodic trajectory. For $-\frac{\Omega}{8} < \delta_1 < \frac{\Omega}{8}$, A and B diverges exponentially and we have an instability while $|\delta_1| > \frac{\Omega}{8}$, A and B are periodic with frequencies $\pm\sqrt{\delta_1^2 - \left(\frac{\Omega}{8}\right)^2}$ giving rise to a quasiperiodic dynamics. For $\mu \neq 0$, the divergence zone changes. The instability zones of the above equations can be worked out analytically only in a perturbative scheme for $\mu < 1$. Accordingly from an inspection of Eqs. (4.3a) and (4.3b) from a self consistency perspective, we try out a solution of the form

$$A(\tau) = \alpha_1\cos\left(\frac{\tau}{2}\right) + \beta_1\sin\left(\frac{\tau}{2}\right). \quad (4.4a)$$

$$B(\tau) = \alpha_2\cos\left(\frac{\tau}{2}\right) + \beta_2\sin\left(\frac{\tau}{2}\right). \quad (4.4b)$$

where α_1 , β_1 , α_2 and β_2 are functions of the time variable τ . Inserting the above in Eqs. (4.3a) and (4.3b) and equating coefficients of $\cos\left(\frac{\tau}{2}\right)$ and $\sin\left(\frac{\tau}{2}\right)$, we get

$$\Delta\dot{\alpha}_1 + \left(\frac{\Delta}{2} + \frac{\mu}{16}\right)\beta_1 - \left[\frac{\delta_1}{\Omega} - \frac{1}{8}\left(1 + \frac{\mu}{2}\right)\right]\alpha_2 = 0. \quad (4.5a)$$

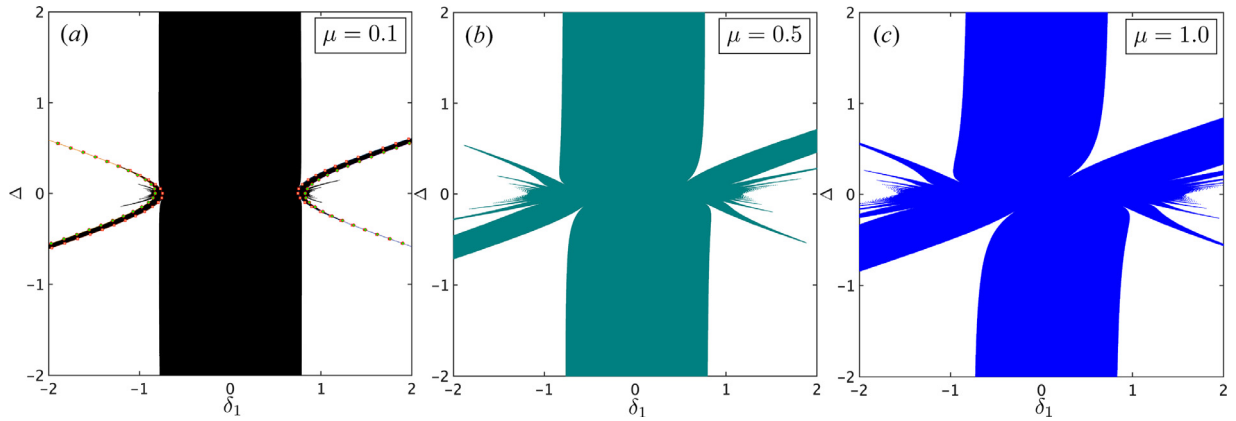


Fig. 3. Zones of stable and unstable region obtained by numerically integrating the slow flow Eqs. (4.3a) and (4.3b) for (a) $\mu = 0.1$, (b) $\mu = 0.5$, and (c) $\mu = 1$. For $\mu = 0.1$, dotted boundary lines are plotted from the analytical solution of Eq. (4.6). Shaded zones correspond to unstable regions, whereas white regions are stable. The instability zones of V obtained from Eqs. (4.12a)–(4.12c) for $\mu = 1$ agree with blue region of the above figure. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$\Delta \dot{\beta}_1 - \left(\frac{\Delta}{2} - \frac{\mu}{16} \right) \alpha_1 - \left[\frac{\delta_1}{\Omega} - \frac{1}{8} \left(1 - \frac{\mu}{2} \right) \right] \beta_2 = 0. \quad (4.5b)$$

$$\Delta \dot{\alpha}_2 + \left(\frac{\Delta}{2} - \frac{\mu}{16} \right) \beta_2 + \left[\frac{\delta_1}{\Omega} + \frac{1}{8} \left(1 + \frac{\mu}{2} \right) \right] \alpha_1 = 0. \quad (4.5c)$$

$$\Delta \dot{\beta}_2 - \left(\frac{\Delta}{2} + \frac{\mu}{16} \right) \alpha_2 + \left[\frac{\delta_1}{\Omega} - \frac{1}{8} \left(1 - \frac{\mu}{2} \right) \right] \beta_1 = 0. \quad (4.5d)$$

The above set of equations have the form $\dot{X}_i = M_{ij}X_j$. The border between stability and instability is obtained from the condition that matrix M has a zero eigenvalue. This gives the conditions

$$\frac{\delta_1}{\Omega} = \begin{cases} -\frac{1}{16} \left[\sqrt{(\mu - 8\Delta)^2 + 4} + \mu \right], \\ \frac{1}{16} \left[\sqrt{(\mu - 8\Delta)^2 + 4} - \mu \right], \\ -\frac{1}{16} \left[\sqrt{(\mu + 8\Delta)^2 + 4} - \mu \right], \\ \frac{1}{16} \left[\sqrt{(\mu + 8\Delta)^2 + 4} + \mu \right]. \end{cases} \quad (4.6)$$

The above boundaries for $\mu = 0.1$ have been plotted in Fig. 3(a). For larger values of μ , this analysis is not appropriate.

The flow equations shown in Eqs. (4.3a) and (4.3b) have to be numerically integrated to get accurate answers, for all values of μ . We show the result with δ_1 along the x -axis and Δ (the quasiperiodicity causing detuning parameter) along the y -axis. Each plot corresponds to a different value of μ . We have considered three values of μ , namely $\mu = 0.1$, 0.5 and 1 . The instability zones are shown in Fig. 3. It should be noted that Fig. 3(a)–(c) correspond to the perturbation theory results and are independent of ϵ for different values of μ . What these figures establish is that the instability zones of the mean and variance coincide in perturbation theory. We will present the perturbation theory for the variance below.

Turning to the dynamics of V and inserting Eq. (4.1) for the frequency ω in Eq. (2.7) we obtain

$$\begin{aligned} \frac{d^3 V}{dt^3} + \Omega^2 \frac{dV}{dt} + 4\Omega\epsilon\delta_1 \frac{dV}{dt} &= -\epsilon\Omega^2 [\cos(\Omega t) + \mu \cos \Omega(1 + \epsilon\Delta)t] \frac{dV}{dt} \\ &+ \frac{\epsilon\Omega^3}{2} [\sin(\Omega t) + \mu(1 + \epsilon\Delta) \sin \Omega(1 + \epsilon\Delta)t] V. \end{aligned} \quad (4.7)$$

We will approach these again in the manner of Section 3 and for $\epsilon = 0$, we write

$$V_0 = A + B \cos(\Omega t) + C \sin(\Omega t), \quad (4.8)$$

where A , B and C are constants. In the Krylov-Bogoliubov spirit, we now make A , B and C slowly varying in time (i.e. $\dot{A}$, $\dot{B}$ and $\dot{C}$ are of $\mathcal{O}(\epsilon)$) and proceeding exactly as in the previous section, we have correct to lowest order in ϵ , for terms on the left hand side of Eq. (4.7)

$$\ddot{V} = -3\Omega^2 \dot{A} \cos(\Omega t) - 3\Omega^2 \dot{B} \sin(\Omega t) + \Omega^3 A \sin(\Omega t) - \Omega^3 B \cos(\Omega t). \quad (4.9a)$$

$$\Omega^2 \dot{V} = \Omega^2 \dot{C} + \Omega^2 \dot{A} \cos(\Omega t) + \Omega^2 \dot{B} \sin(\Omega t) - \Omega^3 A \sin(\Omega t) + \Omega^3 B \cos(\Omega t). \quad (4.9b)$$

$$4\Omega\delta_1\epsilon\dot{V} = 4\Omega\delta_1\epsilon(-\Omega A \sin(\Omega t) + \Omega B \cos(\Omega t)). \quad (4.9c)$$

This yields

$$\ddot{V} + \Omega^2 \dot{V} + 4\Omega\epsilon\delta_1\dot{V} = \Omega^2 \dot{C} - 2\Omega^2 \dot{A} \cos(\Omega t) - 2\Omega^2 \dot{B} \sin(\Omega t) - 4\Omega\epsilon\delta_1 A \sin(\Omega t) + 4\Omega\epsilon\delta_1 B \cos(\Omega t). \quad (4.10)$$

For the *r.h.s.* of Eq. (4.2) evaluation to $\mathcal{O}(\epsilon)$ simply requires using V_0 and $\dot{V}_0$ in the *r.h.s.* of Eq. (4.2) and keeping the terms which have the same structure as on the *r.h.s.* of Eq. (4.3a). We finally arrive at

$$\begin{aligned} \dot{C} - 2\dot{A} \cos(\Omega t) - 2\dot{B} \sin(\Omega t) - \frac{4\delta_1}{\Omega} A \sin(\Omega t) + \frac{4\delta_1}{\Omega} B \cos(\Omega t) - \frac{\epsilon\Omega}{4} [B + \mu A \sin(\epsilon\Delta\Omega t) \\ + \mu B \cos(\epsilon\Delta\Omega t) - 2\mu C \sin(\Omega t)(1 + \mu \cos(\epsilon\Delta\Omega t)) - 2\mu C \cos(\Omega t) \sin(\epsilon\Delta\Omega t)]. \end{aligned} \quad (4.11)$$

Defining $\tau = \epsilon\Delta\Omega t$, we have correct to $\mathcal{O}(\epsilon)$

$$\Delta \frac{dC}{d\tau} = -\frac{B}{4} - \frac{\mu A}{4} \sin(\tau) - \frac{\mu A}{4} \cos(\tau). \quad (4.12a)$$

$$\Delta \frac{dA}{d\tau} = \frac{2\delta_1 B}{\Omega} - \frac{\mu C}{4} \sin(\tau). \quad (4.12b)$$

$$\Delta \frac{dB}{d\tau} = -\frac{2\delta_1 A}{\Omega} - \frac{\mu C}{4} \cos(\tau) - \frac{C}{4}. \quad (4.12c)$$

The perturbative approach to solving these equations are extremely long-winded and given in the Appendix A for formal comparison with Eqs. (4.4)–(4.6)

Instability zones for $\mu = 0.1, 0.5$ and 1 , for the variance V from Eqs. (4.12a)–(4.12c) are shown in Fig. 3(a)–(c). They agree exactly with the corresponding results for the mean. Thus, we have demonstrated in this section that a *Krylov-Bogoliubov* analytic approximation leads to identical instability zones for the mean and variance. We end this section by showing a real time plot of Eqs. (4.3) and (4.12) at one of the edges of bound-unbounded region in the $\Delta - \delta_1$ parameter space. We keep μ, ϵ and Δ fixed and vary δ_1 . One sees from Fig. 4, that for δ_1 slightly on one side of a critical value, both the mean and variance oscillate, while slightly on the other side both diverge.

5. Numerical analysis

We have used fourth-order *Runge-Kutta* (RK4) method (see e.g. [21]) to solve two first order coupled non-autonomous differential equations. Discretization of first order ODE has the form $y_{i+1} - y_i = h\phi(y_i, x_i, h)$, where h is the step size and ϕ is the increment function of $f(x, y)$ in the interval $x_{i+1} \geq x \geq x_i$. Initially, at $\tau = 0$ we choose the initial value of $y_{\tau=0}$ as 0.001 . We are looking at the values of these functions at $\tau = 7000$ with step size $h = 0.001$. Stable oscillatory solution gives with amplitudes with $\sim 10^{-2}$ in the white region, whereas, functional values $y_{\tau} \geq 10^1$ corresponds to divergent solution. Numerical solutions of y_{τ} is shown in the Figs. 4 and 5. All the values of the initial conditions that have used to solve the equations are 0.001 . Other initial values and h do not affect the stability chart diagram.

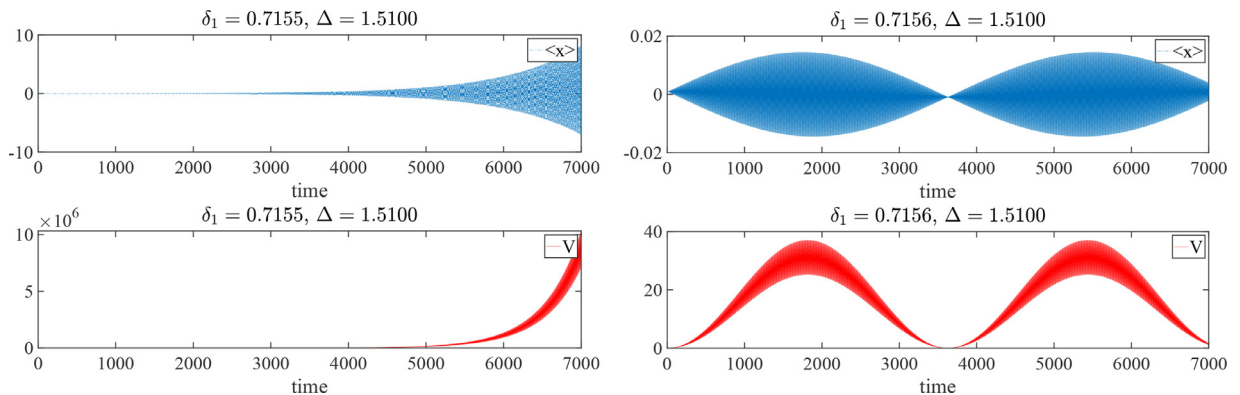


Fig. 4. Numerical section of Eqs. (4.3) and (4.12) with $\mu = 1$, $\Omega = 2\pi$ and for fixed value of $\Delta = 1.51$. Initial values of the corresponding variables in the equations have been taken 0.001 for all the variables. For $\delta = 0.7155$ solutions are diverging for both slow flow of *Mathieu* and variance equation, whereas for $\delta = 0.7156$ both solutions are oscillatory.

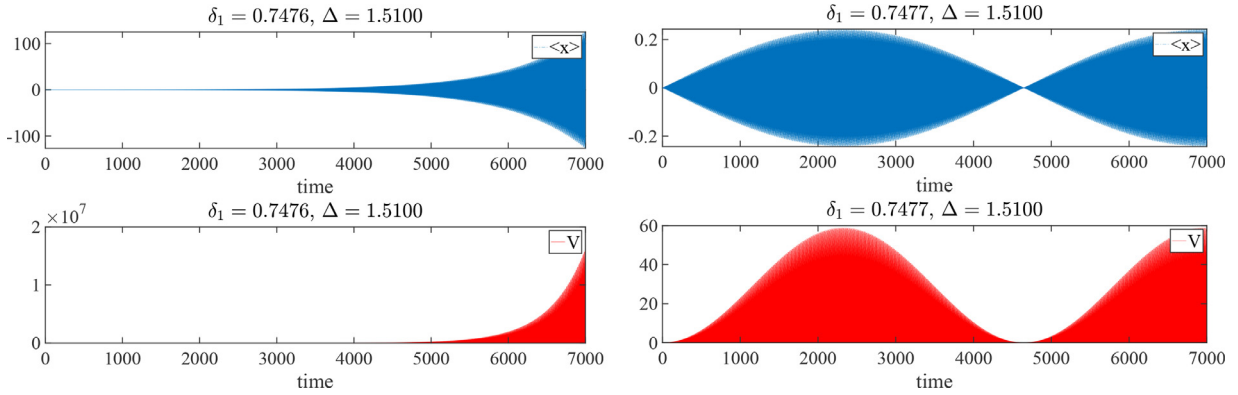


Fig. 5. Numerical solution of Eqs. (2.5) and (2.7) with $\mu = 1$, $\epsilon = 0.1$, $\Omega = 2\pi$ and for fixed value of $\Delta = \sqrt{2} + 0.1$. Initial values of the corresponding variables in the equations have been taken 0.001 for all the variables. For $\delta = 0.7476$ solutions are diverging for both *Mathieu* and variance, whereas for $\delta = 0.7477$ both solutions are oscillatory.

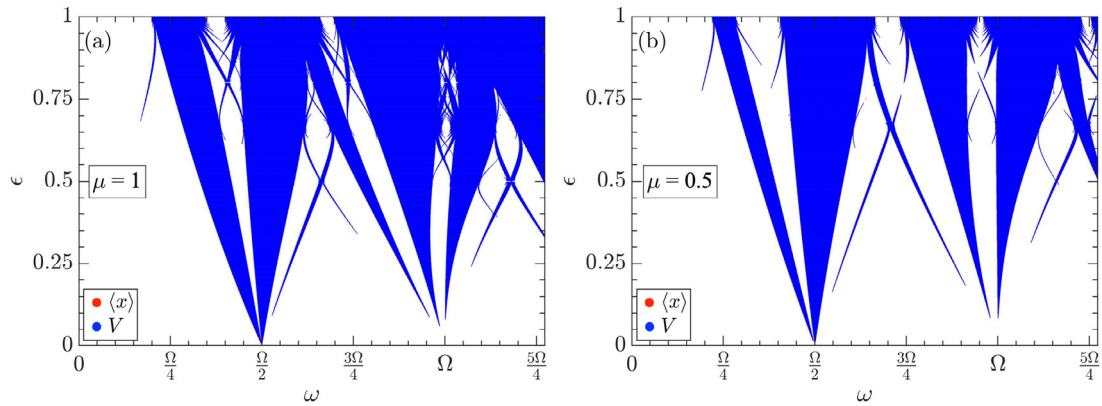


Fig. 6. Numerical solution of Eqs. (2.5) and (2.7) for two different μ , (a) $\mu = 1.0$ and (b) $\mu = 0.5$, with fixed values of $\Delta = -0.5$, $\Omega = 2\pi$ in the $\epsilon - \omega$ parameter space. Shared regions are the unstable solution and white region corresponds to stable solution of the Variance equation. As average and variance solution share the same zones, red and blue colour overlaps each other, thus red colour is not visible in these figures. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

We begin by showing a real time plot of the full equations (Eqs. (2.5) and (2.7)) of motion in Fig. 5. We set $\Omega = 2\pi$, $\mu = 1$, $\epsilon = 0.1$ and Δ an irrational number $\sqrt{2} + 0.1 \simeq 1.51$ (the same values as in Fig. 4 except ϵ , as it is implicit in the Eqs. (4.3) and (4.12)). To confirm mean $\langle x \rangle$ and the variance V share the same boundary as shown in the Fig. 3, we choose δ_1 slightly lower than a critical value δ_c shows oscillatory behaviour, however, a driving behaviour for δ_1 slightly greater than δ_c . The same behaviour can be observed in Fig. 4 - the primary difference between Fig. 4 (perturbation analysis) and Fig. 5 (complete numerical analysis) is that the critical value of δ_1 in the approximate perturbation analysis is 0.7156, while for the exact numerical analysis it is 0.7477. The difference in critical value on basis of perturbation analysis and exact numerical result is expected but the phenomenon can not change.

We now show the overlapping resonance zones of Eqs. (2.5) and (2.7) obtained from a numerical analysis. Our results from Sections 2-4 can be summarized as follows: A parametric forcing of period T can induce responses of period $2T$ and T in Eq. (2.5) and periods $3T$, $2T$ and T in Eq. (2.7). Resonances and hence potential instabilities can exist when the natural frequency matches the frequency of these responses. For Eq. (2.5), where the natural frequency is ω , the possible resonance conditions are $\omega = \frac{\Omega}{2}$ and $\omega = \Omega$ (weaker resonances will occur at $(2n+1)\frac{\Omega}{2}$ and $2n\Omega$, $n = 1, 2, \dots$). For Eq. (2.7), the natural frequency is 2ω and instabilities are possible for the resonance conditions $2\omega = \frac{2\pi}{3T}$, $\frac{2\pi}{2T}$, $\frac{2\pi}{T}$ i.e. $\omega = \frac{\Omega}{6}$, $\frac{\Omega}{4}$ and $\frac{\Omega}{2}$. We argued at the end of the Sec. [2], that there will be no instability at $\frac{\Omega}{6}$ and showed explicitly in Sec. [3], that there is no instability around $\frac{\Omega}{4}$. Consequently it is at $\omega = \frac{\Omega}{2}$, we have the first instability in both $\langle x \rangle$ and V . Weaker instabilities occur around $\omega = \Omega$, $\frac{3\Omega}{2}$, $2\Omega \dots$ etc. Our numerics show no instability before $\omega = \frac{\Omega}{2} \pm \delta_1$.

The instability zones for $\langle x \rangle$ in the vicinity of $\frac{\Omega}{2}$ have already been shown in Fig. 2(a) and (b) in the $\omega - \epsilon$ plane. We now show the instability zones for V for the same parameter values in Fig. 6(a) and (b) with the zones of Fig. 2(a) and (b) for $\langle x \rangle$ superposed. It is obvious that zones overlap very well. For completeness, we now present results in the $\Delta - \omega$ plane and in Fig. 7 and 8(a) and (b) show the overlap of instability zones of $\langle x \rangle$ and V for two different values of μ namely $\mu = 0.5$ and

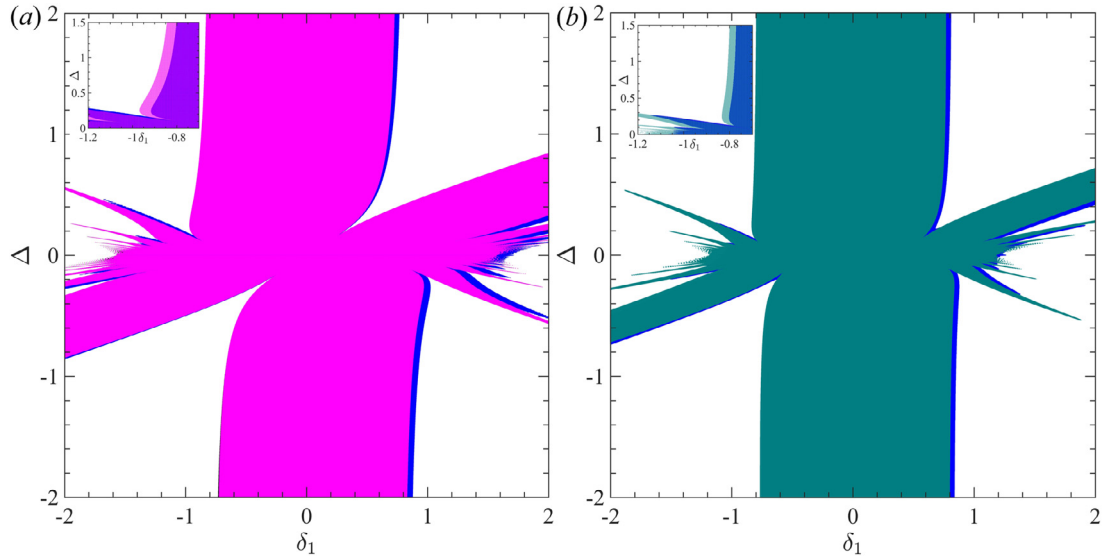


Fig. 7. A comparison with the instability zones coming from the exact Mathieu equation i.e. Eq. (2.7) (magenta) and slow flow of Mathieu equation i.e. Eq. (4.3a) and Eq. (4.3b) (blue) for $\mu = 1$, $\epsilon = 0.1$, $\Omega = 2\pi$ shown in panel (a). Similarly variance Eq. (2.11) (teal) and variance slow flow Eq. (4.12a) - Eq. (4.12c) (blue) for $\mu = 0.5$, $\epsilon = 0.1$, $\Omega = 2\pi$, shown in panel (b). In both (a) and (b) show how the full equation when plotted for definite but small ϵ show small deviations from the approximation but the result that the instability zones coincide hold. Inset is blow up of the right side of the unstable region to show shifting portion of the region below.

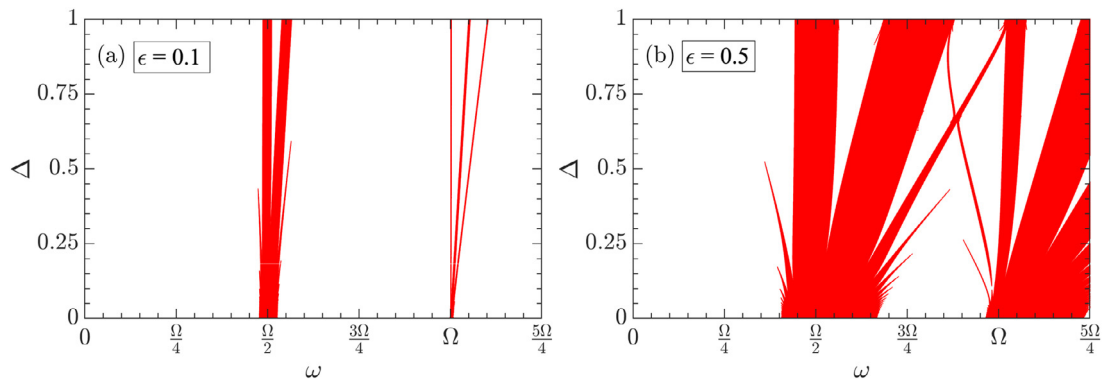


Fig. 8. Numerical solution of Eq. (2.5) (valid for Eq. (2.7) also) for two different ϵ , (a) $\epsilon = 0.1$ and (b) $\epsilon = 0.5$, with fixed values of $\mu = 1.0$, $\Omega = 2\pi$ in the $\Delta - \omega$ parameter space. Red shaded region is the unstable solution and white region corresponds to stable solution of the Mathieu (Variance) equation.

$\mu = 1.0$. Taken in conjunction with the overlap zones shown in Fig. 1, we assert that in a quasiperiodically forced quantum Mathieu oscillator the resonance zones of the mean and Variance are identical.

6. Conclusion

We have considered a quasiperiodically driven quantum parametric oscillator near the 2: 2: 1 resonance which produces the most complicated instability pattern in the three-dimensional space spanned by ϵ , Δ and μ . Here ϵ and $\epsilon\mu$ are the amplitudes of the quasiperiodic drives and Δ is the detuning parameter for the quasiperiodicity. We find that in the entire three-dimensional space, the instability zones of the mean position and the variance remain the same as they do in the two dimensional (frequency and amplitude) space of the periodically driven system. We establish this from perturbation theory and accurate numerical solution.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

In this Appendix, we show explicitly how the amplitude equation for the variance V as given in Eqs. (4.12a)–(4.12c) can be handled in a manner identical to our treatment of Eqs. (4.4a) and (4.4b)– the amplitude equations for the mean. We begin by setting $\mu = 0$, which leads to

$$\Delta \dot{C} = -\frac{B}{4}, \quad (\text{A.1a})$$

$$\Delta \dot{A} = \frac{2\delta_1 B}{\Omega}, \quad (\text{A.1b})$$

$$\Delta \dot{B} = -\frac{2\delta_1 A}{\Omega} - \frac{C}{4}. \quad (\text{A.1c})$$

The solutions are going to be of the form $e^{\lambda\tau/\Delta}$, where the growth rates λ are obtained from the solvability condition

$$\text{Det} \begin{pmatrix} \lambda & 0 & \frac{1}{4} \\ 0 & \lambda & -\frac{2\delta_1}{\Omega} \\ \frac{1}{4} & \frac{2\delta_1}{\Omega} & \lambda \end{pmatrix} = 0, \quad (\text{A.2})$$

leading to

$$\lambda \left[(\lambda)^2 + \left(\frac{2\delta_1}{\Omega} \right)^2 \right] - \frac{\lambda}{16} = 0. \quad (\text{A.3})$$

This leads to the non-zero growth rates of $\lambda = \pm \sqrt{\frac{1}{16} - \left(\frac{2\delta_1}{\Omega} \right)^2}$. For $|\frac{\delta_1}{\Omega}| < \frac{1}{8}$, we have real values of λ and that means the amplitudes (A, B, C) will grow exponentially in time and there will be an instability. For $|\frac{\delta_1}{\Omega}| > \frac{1}{8}$, the solutions are oscillatory and hence bounded. The overall dynamics is quasiperiodic since the solutions are of the form $V = C(t) + A(t) \cos \Omega t + \sin \Omega t$ and we have superposition of the frequencies $\frac{\Omega}{2} + \frac{\epsilon\Omega}{8}$ and $\frac{\Omega}{2} - \frac{\epsilon\Omega}{8}$. We note that, this is in exact agreement with Eq. (4.6) (which gives the boundaries for the mean) when $\mu = 0$ and $\Delta \rightarrow 0$. We now go a little further and include μ and Δ in a nontrivial manner.

With $\mu = 0$, the solution to Eqs. (4.12a)–(4.12c) has a constant part (this corresponds to the $\lambda = 0$ case) and hence for $\mu \neq 0$ but very small, we expect a slowly varying solution with no oscillatory part. Having noted that we can inspection, write down the trial solutions

$$A(\tau) = \alpha_1(\tau) \cos(\tau) + \beta_1(\tau) \sin(\tau) + \gamma_1(\tau). \quad (\text{A.4a})$$

$$B(\tau) = \alpha_2(\tau) \cos(\tau) + \beta_2(\tau) \sin(\tau) + \gamma_2(\tau). \quad (\text{A.4b})$$

$$C(\tau) = \alpha_3(\tau) \cos(\tau) + \beta_3(\tau) \sin(\tau) + \gamma_3(\tau). \quad (\text{A.4c})$$

Inserting these solutions in Eqs. (4.12a)–(4.12c), we obtain in the following set of nine equations

$$\Delta \dot{\alpha}_1 + \Delta \beta_1 - \frac{2\delta_1}{\Omega} \alpha_2 = 0, \quad (\text{A.5a})$$

$$\Delta \dot{\beta}_1 - \Delta \alpha_1 - \frac{2\delta_1}{\Omega} \beta_2 + \frac{\mu}{4} \gamma_3 = 0, \quad (\text{A.5b})$$

$$\Delta \dot{\gamma}_1 - \frac{2\delta_1}{\Omega} \gamma_2 + \frac{\mu}{8} \beta_3 = 0, \quad (\text{A.5c})$$

$$\Delta \dot{\alpha}_2 + \Delta \beta_2 + \frac{2\delta_1}{\Omega} \alpha_1 + \frac{\alpha_3}{4} = 0, \quad (\text{A.5d})$$

$$\Delta \dot{\beta}_2 - \Delta \alpha_2 + \frac{2\delta_1}{\Omega} \beta_1 + \frac{\beta_3}{4} + \frac{\mu}{4} \gamma_3 = 0, \quad (\text{A.5e})$$

$$\Delta \dot{\gamma}_2 + \frac{2\delta_1}{\Omega} \gamma_1 + \frac{\gamma_3}{4} + \frac{\mu}{4} \beta_3 = 0, \quad (\text{A.5f})$$

$$\Delta \dot{\alpha}_3 + \Delta \beta_3 + \frac{\alpha_2}{4} + \frac{\mu}{4} \gamma_2 = 0, \quad (\text{A.5g})$$

$$\Delta \dot{\beta}_3 - \Delta \alpha_3 + \frac{\beta_2}{4} + \frac{\mu}{4} \gamma_1 = 0, \quad (\text{A.5h})$$

$$\Delta \dot{\gamma}_3 + \frac{\gamma_2}{4} + \frac{\mu}{8} \alpha_2 + \frac{\mu}{8} \beta_1 = 0, \quad (\text{A.5i})$$

To find the growth rates of the coefficients $\alpha_1, \beta_1, \gamma_1, \dots$ etc., we need to find the eigenvalues of the 9×9 matrix

$$\begin{pmatrix} 0 & \Delta & -\frac{2\delta_1}{\Omega} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\Delta & 0 & 0 & -\frac{2\delta_1}{\Omega} & 0 & 0 & 0 & 0 & \frac{\mu}{4} \\ \frac{2\delta_1}{\Omega} & 0 & 0 & \Delta & \frac{1}{4} & 0 & 0 & 0 & \frac{\mu}{4} \\ 0 & \frac{2\delta_1}{\Omega} & -\Delta & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 & 0 & \Delta & 0 & \frac{\mu}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{4} & -\Delta & 0 & \frac{\mu}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\mu}{8} & 0 & -\frac{2\delta_1}{\Omega} & 0 \\ 0 & 0 & 0 & 0 & \frac{\mu}{8} & 0 & \frac{2\delta_1}{\Omega} & 0 & \frac{1}{4} \\ 0 & \frac{\mu}{8} & \frac{\mu}{8} & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 \end{pmatrix}$$

For $\mu = 0$, the marginal stability situation leads to the result

$$\Delta^2 + 4 \left(\frac{1}{64} - \frac{\delta_1^2}{\Omega^2} \right) = 0, \quad (\text{A.6})$$

which is in exact agreement with the $\mu \rightarrow 0$ result of Eq. (4.6). It is quite remarkable that the 4×4 matrix of Eqs. (4.5a)–(4.5d) for the dynamics of the mean and the 9×9 matrix of Eqs. (A.5a)–(A.5i) for the variance give the identical condition for a zero eigenvalue (the marginal stability case) in the limit of $\mu \rightarrow 0$. Diagonalization of the 9×9 matrix for $\mu \neq 0$ was not possible to do analytically. However, for $\mu < 1$ and Δ also small so that $\mu\Delta$ can be ignored, we find that a perturbative correction to the condition of zero eigenvalue at $\mathcal{O}(\mu)$ is same in both cases. We have not found a way of demonstrating analytic overlap beyond this.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.cnsns.2020.105537](https://doi.org/10.1016/j.cnsns.2020.105537).

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